

Introduction and Establishment of *Aedes albopictus* (Diptera: Culicidae) in Managua, Nicaragua.

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Aedes aegypti (L.) is the main vector of dengue virus and more recently chikungunya virus in Latin America. However, the Asian tiger mosquito *Aedes albopictus* (Skuse, 1894) is expanding its global range and increasing its role in transmission of these diseases. In this report, we suggest that *Ae. albopictus* was introduced to the Department of Managua, Nicaragua, in 2010 via two independent routes and demonstrate its dissemination and establishment in urban neighborhoods by 2012. The coexistence of two competent vector species could alter the epidemiology of dengue and chikungunya as well as indicate the need for new strategies aimed at vector control.

Salinity tolerant *Aedes aegypti* and *Ae. albopictus*-Infection with dengue virus and contribution to dengue transmission in a coastal peninsula.

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Aedes aegypti and *Ae. albopictus* are major arboviral vectors that are considered to lay eggs, and undergo preimaginal development only in fresh water collections. However, recently they have been also shown to develop in coastal brackish water habitats. The ability of the biologically variant salinity-tolerant *Aedes* vectors to transmit arboviral diseases is unknown. We therefore, investigated the infection of salinity-tolerant *Aedes* mosquitoes with dengue virus (DENV) and analysed dengue incidence and rainfall data to assess the contribution of salinity-tolerant *Aedes* vectors to dengue transmission in the coastal Jaffna peninsula in Sri Lanka. Brackish and fresh water developing female *Ae. aegypti* and *Ae. albopictus* were tested for their ability to become infected with DENV through in vitro blood feeding and then transmit DENV vertically to their progeny. An immunochromatographic test for the NS1 antigen was used to detect DENV. Temporal variation in dengue incidence in relation to rainfall was analysed for the peninsula and other parts of Sri Lanka. *Aedes aegypti* and *Ae. albopictus* developing in brackish water, became infected with DENV through in vitro blood feeding and the infected mosquitoes were able to vertically transmit DENV to their progeny. Monsoonal rainfall was the discernible factor responsible for the seasonal increase in dengue incidence in the peninsula and elsewhere in Sri Lanka. Fresh water *Aedes* vectors are main contributors to the increased dengue incidence that typically follows monsoons in the Jaffna peninsula and elsewhere in Sri Lanka. It is possible however, that brackish water-developing *Aedes* constitute a perennial reservoir for DENV to maintain a basal level of dengue transmission in coastal areas of the peninsula during the dry season, and this supports increased transmission when monsoonal rains expand populations of fresh water *Aedes*.

A Comprehensive Entomological, Serological and Molecular Study of 2013 Dengue Outbreak of Swat, Khyber Pakhtunkhwa, Pakistan.

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Aedes aegypti and *Aedes albopictus* play a fundamental role in transmission of dengue virus to humans. A single infected *Aedes* mosquito is capable to act as a reservoir/amplifier host for dengue virus and may cause epidemics via horizontal and vertical modes of dengue virus (DENV) transmission. The present and

future dengue development can be clarified by understanding the elements which help the dissemination of dengue transmission. The current study deals with molecular surveillance of dengue in addition to ecological and social context of 2013 dengue epidemics in Swat, Pakistan. Herein, we reported dengue vectors surveillance in domestic and peridomestic containers in public and private places in 7 dengue epidemic-prone sites in District Swat, Pakistan from July to November 2013. Using the Flaviviruses genus-specific reverse transcriptase (RT) semi nested-PCR assay, we screened blood samples (N = 500) of dengue positive patients, 150 adult mosquito pools and 25 larval pools. The 34 adult and 7 larval mosquito pools were found positive. The adult positive pools comprised 30 pools of *Ae. aegypti* and 4 pools of *Ae. albopictus*, while among the 7 larval pools, 5 pools of *Ae. aegypti* and 2 pools of *Ae. albopictus* were positive. The detected putative genomes of dengue virus were of DENV-2 (35% in 14 mosquito pools & 39% in serum) and DENV-3 (65% in 27 mosquito pools & 61% in serum). The higher vector density and dengue transmission rate was recorded in July and August (due to favorable conditions for vector growth). About 37% of *Ae. aegypti* and 34% *Ae. albopictus* mosquitoes were collected from stagnant water in drums, followed by drinking water tanks (23% & 26%), tires (20% & 18%) and discarded containers (10% & 6%). Among the surveyed areas, Saidu was heavily affected (26%) by dengue followed by Kanju (20%) and Landikas (12%). The maximum infection was observed in the age group of 45 (25%) years and was more in males (55.3%) as compared to females (44.7%). The increase in vector mosquito density and the subsequent viral transmission was determined by a complex interplay of ecological, biological and social factors. The suitable environmental conditions and discriminable role of *Aedes* through trans-ovarial transmission of DENV is indispensable in the recent geographic increase of dengue in Pakistan. Climate change affects the survival and dispersion of vectors as well as the transmission rates of dengue. Control of *Aedes* mosquitoes (vectors) and elimination of breeding sources must be emphasized and prioritized. Such actions may not only reduce the risk of dengue transmission during epidemics, but also minimize the chances of dengue viruses establishment in new (non endemic) areas of the region.

Can Wolbachia be used to control malaria?

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Malaria is a mosquito-borne infectious disease caused by *Plasmodium* parasites transmitted by the infectious bite of *Anopheles* mosquitoes. Vector control of malaria has predominantly focused on targeting the adult mosquito through insecticides and bed nets. However, current vector control methods are often not sustainable for long periods so alternative methods are needed. A novel biocontrol approach for mosquito-borne diseases has recently been proposed, it uses maternally inherited endosymbiotic *Wolbachia* bacteria transinfected into mosquitoes in order to interfere with pathogen transmission. Transinfected *Wolbachia* strains in *Aedes aegypti* mosquitoes, the primary vector of dengue fever, directly inhibit pathogen replication, including *Plasmodium gallinaceum*, and also affect mosquito reproduction to allow *Wolbachia* to spread through mosquito populations. In addition, transient *Wolbachia* infections in *Anopheles gambiae* significantly reduce *Plasmodium* levels. Here we review the prospects of using a *Wolbachia*-based approach to reduce human malaria transmission through transinfection of *Anopheles* mosquitoes.

Dengue Fever in mainland China.

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Dengue is an acute emerging infectious disease transmitted by *Aedes* mosquitoes and has become a serious global public health problem. In mainland China, a number of large dengue outbreaks with serious consequences have been reported as early as 1978. In the three decades from 1978 to 2008, a total of 655,324 cases were reported, resulting in 610 deaths. Since the 1990s, dengue epidemics have spread gradually from Guangdong, Hainan, and Guangxi provinces in the southern coastal regions to the relatively northern and western regions including Fujian, Zhejiang, and Yunnan provinces. As the major transmission vectors of dengue viruses, the biological behavior and vectorial capacity of *Aedes* mosquitoes have undergone significant changes in the last two decades in mainland China, most likely the result of urbanization and global climate changes. In this review, we summarize the geographic and temporal distributions, the serotype and genotype distributions of dengue viruses in mainland China, and analyze the current status of surveillance and control of vectors for dengue transmission.

Dengue fever in international travelers.

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Dengue virus infection is becoming increasingly recognized as one of the world's major emerging infectious diseases. Although only a few systematic studies have been conducted to assess the incidence and clinical course of dengue fever in travelers, it is now possible to estimate risk factors for travelers to areas of endemicity. Dengue virus and its vector, *Aedes* mosquitoes, benefit from human habitation and travel-related aspects of human behavior. Thus, travelers serve an important double role as potential victims of the disease and as vehicles for further spread of dengue.

Global cross-talk of genes of the mosquito *Aedes aegypti* in response to dengue virus infection.

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The mosquito *Aedes aegypti* is the primary vector of dengue virus (DENV) infection in humans, and DENV is the most important arbovirus across most of the subtropics and tropics worldwide. The early time periods after infection with DENV define critical cellular processes that determine ultimate success or failure of the virus to establish infection in the mosquito. To identify genes involved in these processes, we performed genome-wide transcriptome profiling between susceptible and refractory *A. aegypti* strains at two critical early periods after challenging them with DENV. Genes that responded coordinately to DENV infection in the susceptible strain were largely clustered in one specific expression module, whereas in the refractory strain they were distributed in four distinct modules. The susceptible response module in the global transcriptional network showed significant biased representation with genes related to energy metabolism and DNA replication, whereas the refractory response modules showed biased representation across different metabolism pathway genes including cytochrome P450 and DDT [1,1,1-Trichloro-2,2-bis(4-chlorophenyl) ethane] degradation genes, and genes associated with cell growth and death. A common core set of coordinately expressed genes was observed in both the susceptible and refractory mosquitoes and included genes related to the Wnt (Wnt: wingless [wg] and integration 1 [int1] pathway), MAPK (Mitogen-activated protein kinase), mTOR (mammalian target of rapamycin) and JAK-STAT (Janus Kinase - Signal Transducer and Activator of Transcription) pathways. Our

data revealed extensive transcriptional networks of mosquito genes that are expressed in modular manners in response to DENV infection, and indicated that successfully defending against viral infection requires more elaborate gene networks than hosting the virus. These likely play important roles in the global-cross talk among the mosquito host factors during the critical early DENV infection periods that trigger the appropriate host action in susceptible vs. refractory mosquitoes.

Detection of dengue viruses and Wolbachia in *Aedes aegypti* and *Aedes albopictus* larvae from four urban localities in Kuala Lumpur, Malaysia.

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Dengue fever (DF) is currently one of the most important mosquito-borne diseases that affects humans. Dengue fever (DF) and dengue hemorrhagic fever (DHF) are caused by four serotypes of dengue viruses (DENV-1 to DENV-4). The main vector transmitting dengue is *Aedes aegypti* while *Aedes albopictus* acts as a secondary vector. As treatment is unavailable and the first dengue vaccine approved in Mexico, Dengvaxia® has yet to be accepted worldwide, prevention of the disease relies heavily on surveillance and control of mosquito vectors. A transgene driver, Wolbachia was found to limit the transmission of dengue virus in *Aedes* mosquitoes. Wolbachia alone was able to inhibit viral replication, dissemination and transmission in *A. aegypti* mosquitoes in experimental studies. In *A. albopictus*, Wolbachia did not affect the replication of dengue virus but was able to reduce the viral infection of mosquito salivary glands and limit transmission. Studies on Wolbachia have all been carried out in adult *Aedes* mosquitoes, hence this study was conducted to determine the presence of dengue virus serotypes and Wolbachia in *A. aegypti* and *A. albopictus* larvae collected from ovitraps in four localities in Kuala Lumpur viz. Happy Gardens, IMU Bukit Jalil, Ampang and Taman Yarl. Another objective of this study was to determine the association between dengue virus serotypes and the presence of Wolbachia in *A. aegypti* and *A. albopictus* larvae. A total of 300 mosquito larvae was collected; 99 (Happy Gardens), 85 (Bukit Jalil), 73 (Ampang) and 43 (Taman Yarl). Out of 300 larvae collected, 284 were identified as *A. albopictus* and 16 others were identified as *A. aegypti*. Of the 284 *A. albopictus* larvae collected, 211 (74.3%) and 73 (25.7%) were found to be negative and positive for dengue virus respectively. The dengue serotypes detected were 2 DENV-2 (2.7%), 58 DENV-3 (79.5%) and 13 DENV-4 (17.8%). DENV-1 was not detected in any of the *A. albopictus* larvae. For *A. aegypti*, out of 16 *A. aegypti* larvae collected, 12 (75%) were found to be negative and 4 (25%) were positive for DENV-2. For the detection of Wolbachia in *A. albopictus*, 71 out of 284 (25%) and 213 (75%) larvae were found to be positive and negative for Wolbachia respectively. For *A. aegypti*, 4 (25%) and 12 (75%) out of 16 larvae were positive and negative for Wolbachia respectively. This is the first report of Wolbachia in *A. albopictus* and *A. aegypti* larvae in Malaysia. A chisquare test analysis to determine the association between dengue virus and Wolbachia in *A. albopictus* and *A. aegypti* larvae collected from the four localities in Kuala Lumpur showed that there was no association ($\chi^2 = 3.080$; $df = 1$; $P > 0.05$).

Larval competition alters susceptibility of adult *Aedes* mosquitoes to dengue infection.

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Dengue, the most important human arboviral disease, is transmitted primarily by *Aedes aegypti* and, to a lesser extent, by *Aedes albopictus*. The current distributions of these invasive species overlap and are

affected by interspecific larval competition in their container habitats. Here we report that competition also enhances dengue infection and dissemination rates in one of these two vector species. We determined the effects of competition on adult *A. aegypti* and *A. albopictus*, comparing their susceptibility to infection with a Southeast Asian strain of dengue-2 virus. High levels of intra- or interspecific competition among larvae enhanced the susceptibility of *A. albopictus* to dengue virus infection and potential for transmission, as indicated by disseminated infections. Doubling the number of competing larvae (*A. albopictus* or *A. aegypti*), led to a significant (more than 60%) increase in the proportion of *A. albopictus* with disseminated dengue-2 infection. Competition-enhanced vector competence appears to result from a reduction in 'barriers' (morphological or physiological) to virus infection and dissemination and may contribute to the importance of *A. albopictus* in dengue transmission. Similar results for other unrelated arboviruses suggest that larval competition, common in mosquitoes, should be considered in estimates of vector competence for pathogens that infect humans.

Susceptibility of Indigenous and Transplanted Mosquito Spp. to Dengue Virus in Japan.

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Dengue fever, an acute, mosquito-borne, febrile illness caused by *Flavivirus* spp., is a problem in Africa, South and Southeast Asia, Latin America, and the Caribbean. A dengue outbreak occurred after nearly 70 years of absence or no detection, and then 158 autochthonous cases occurred in Japan from August to October 15, 2014. The most competent mosquito vectors for dengue virus transmission were *Aedes aegypti* and *A. albopictus*. Since *A. albopictus* is widely distributed across Japan and *A. aegypti* recently invaded Japan by airplane, we examined the susceptibility of these species to infection by dengue virus.
